

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics
PRACTICAL - VI

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Practical 6: Decision Making under Uncertainty

Part A: Develop a Decision Tree Model

CODE:-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# 1. Load dataset and display first 10 records
df = pd.read_csv("supermarket_sales.csv")
print("First 10 Records:")
print(df.head(10))

# 2. Display number of records, columns, column names and data types
print("\nNumber of Records:", df.shape[0])
print("Number of Columns:", df.shape[1])
print("\nColumn Names:")
print(df.columns.tolist())
print("\nData Types:")
print(df.dtypes)

# 3. Display all unique values of product_line
print("\nUnique Product Lines:")
print(df["product_line"].unique())

# 4. Calculate total revenue generated by each product line
total_revenue = df.groupby("product_line")["revenue"].sum()
print("\nTotal Revenue by Product Line:")
print(total_revenue)

# 5. Calculate average revenue generated by each product line
average_revenue = df.groupby("product_line")["revenue"].mean()

print("\nAverage Revenue by Product Line:")
print(average_revenue)

# 6. Calculate total quantity sold for each product line
total_quantity = df.groupby("product_line")["quantity"].sum()
print("\nTotal Quantity Sold:")
print(total_quantity)
```

7. Calculate median value of revenue

```
median_revenue = df["revenue"].median()
print("\nMedian Revenue:", median_revenue)
```

8 & 9. Classify transactions as High Revenue or Low Revenue

```
df["revenue_level"] = np.where(
    df["revenue"] >= median_revenue,
    "High Revenue",
    "Low Revenue"
)
print("\nRevenue Classification:")
print(df[["revenue", "revenue_level"]].head(10))
```

10. Number of High and Low Revenue transactions for each product line

```
revenue_counts = pd.crosstab(
    df["product_line"],
    df["revenue_level"]
)
print("\nRevenue Counts:")
print(revenue_counts)
```

11. Calculate probability of High and Low Revenue

```
revenue_probability = revenue_counts.div(
    revenue_counts.sum(axis=1),
    axis=0
)
print("\nRevenue Probabilities:")
print(revenue_probability)
```

12. Verify probabilities add up to 1

```
probability_check = revenue_probability.sum(axis=1)
print("\nProbability Check:")
print(probability_check)
```

13. Average revenue for High and Low Revenue transactions

```
revenue_payoff = df.pivot_table(
    index="product_line",
    columns="revenue_level",
    values="revenue",
    aggfunc="mean"
)
print("\nAverage Revenue / Payoff:")
print(revenue_payoff)
```

14. Create DataFrame containing Product Line,**# Revenue Level, Probability and Payoff**

```
decision_data = []
for product in df["product_line"].unique():
    for level in ["High Revenue", "Low Revenue"]:
        probability = revenue_probability.loc[product, level]
```

```

    payoff = revenue_payoff.loc[product, level]
    decision_data.append([
        product,
        level,
        probability,
        payoff
    ])
decision_df = pd.DataFrame(
    decision_data,
    columns=[
        "Product Line",
        "Revenue Level",
        "Probability",
        "Payoff"
    ]
)
print("\nDecision DataFrame:")
print(decision_df)

```

15. Identify decision alternatives and uncertain outcomes

```

decision_alternatives = df["product_line"].unique()
uncertain_outcomes = ["High Revenue", "Low Revenue"]
print("\nDecision Alternatives:")
print(decision_alternatives)
print("\nUncertain Outcomes:")
print(uncertain_outcomes)

```

16 & 17. Construct and display decision tree

```

print("\nDECISION TREE")
print("=" * 60)
for product in decision_alternatives:
    print("\n" + product)
    print(" |")
    for level in uncertain_outcomes:
        probability = revenue_probability.loc[product, level]
        payoff = revenue_payoff.loc[product, level]

        print(
            f"   +-- {level} "
            f"(Probability = {probability:.3f}, "
            f"Payoff = {payoff:.2f})"
        )

```

18. Visualization showing total revenue

```

plt.figure(figsize=(10, 6))

total_revenue.sort_values().plot(
    kind="barh"
)
plt.title("Total Revenue by Product Line")

```

```
plt.xlabel("Total Revenue")
plt.ylabel("Product Line")
plt.tight_layout()
plt.show()
```

19. Visualization showing probability

of High and Low Revenue

```
revenue_probability.plot(
    kind="bar",
    figsize=(10, 6)
)
plt.title("Probability of High and Low Revenue")
plt.xlabel("Product Line")
plt.ylabel("Probability")
plt.xticks(rotation=45)
plt.legend(title="Revenue Level")
plt.tight_layout()
plt.show()
```

20. Product line with highest probability of High Revenue

```
highest_high_probability_product = (
    revenue_probability["High Revenue"].idxmax()
)
highest_high_probability = (
    revenue_probability["High Revenue"].max()
)
print(
    "\n20. Highest Probability of High Revenue:",
    highest_high_probability_product
)
print(
    "Probability:",
    highest_high_probability
)
print("KHUSHI T032")
```

OUTPUT:-

```
First 10 Records:
  invoice_id branch    city customer_type gender_customer \
0  750-67-8428    A  Yangon      Member      Female
1  226-31-3081    C  Naypyitaw   Normal      Female
2  631-41-3108    A  Yangon      Normal      Male
3  123-19-1176    A  Yangon      Member      Male
4  373-73-7910    A  Yangon      Normal      Male
5  699-14-3026    C  Naypyitaw   Normal      Male
6  355-53-5943    A  Yangon      Member      Female
7  315-22-5665    C  Naypyitaw   Normal      Female
8  665-32-9167    A  Yangon      Member      Female
9  692-92-5582    B  Mandalay    Member      Female

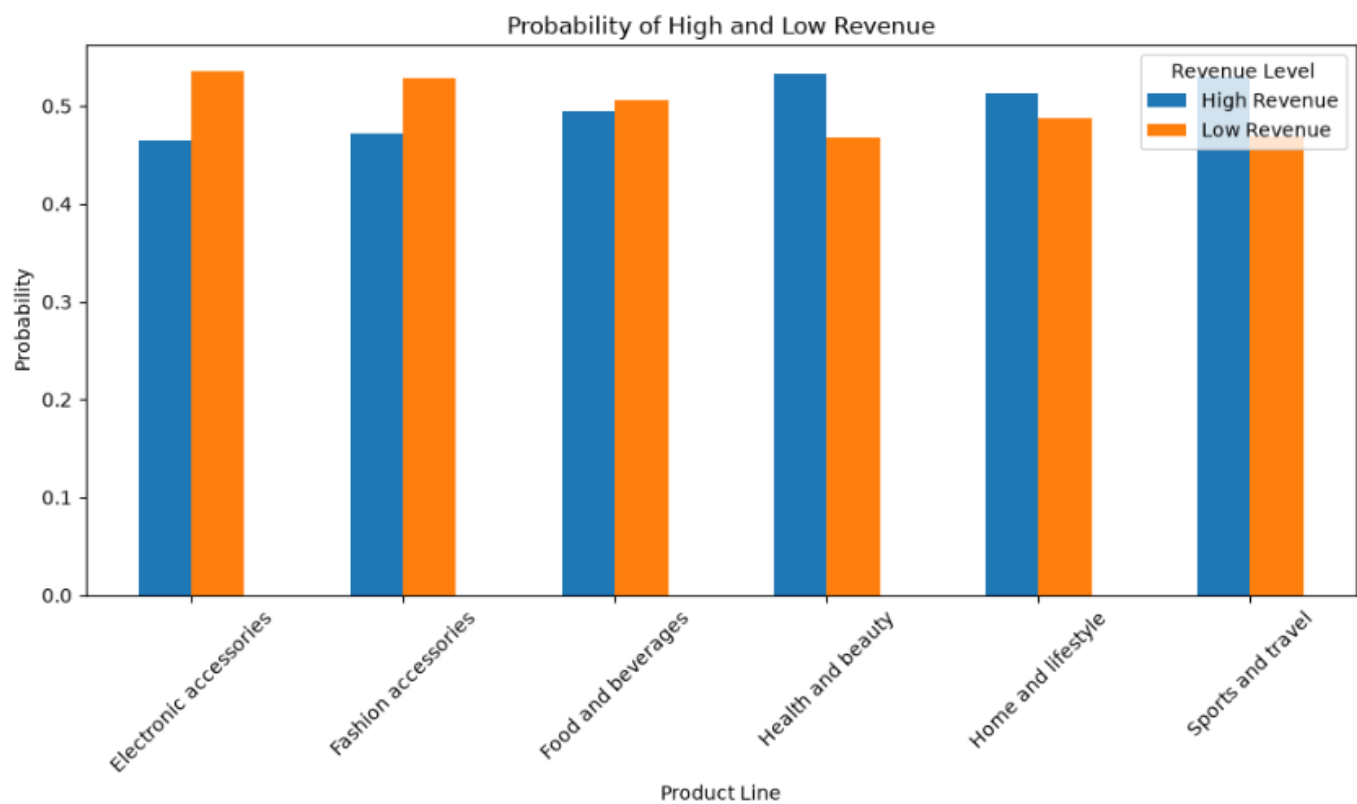
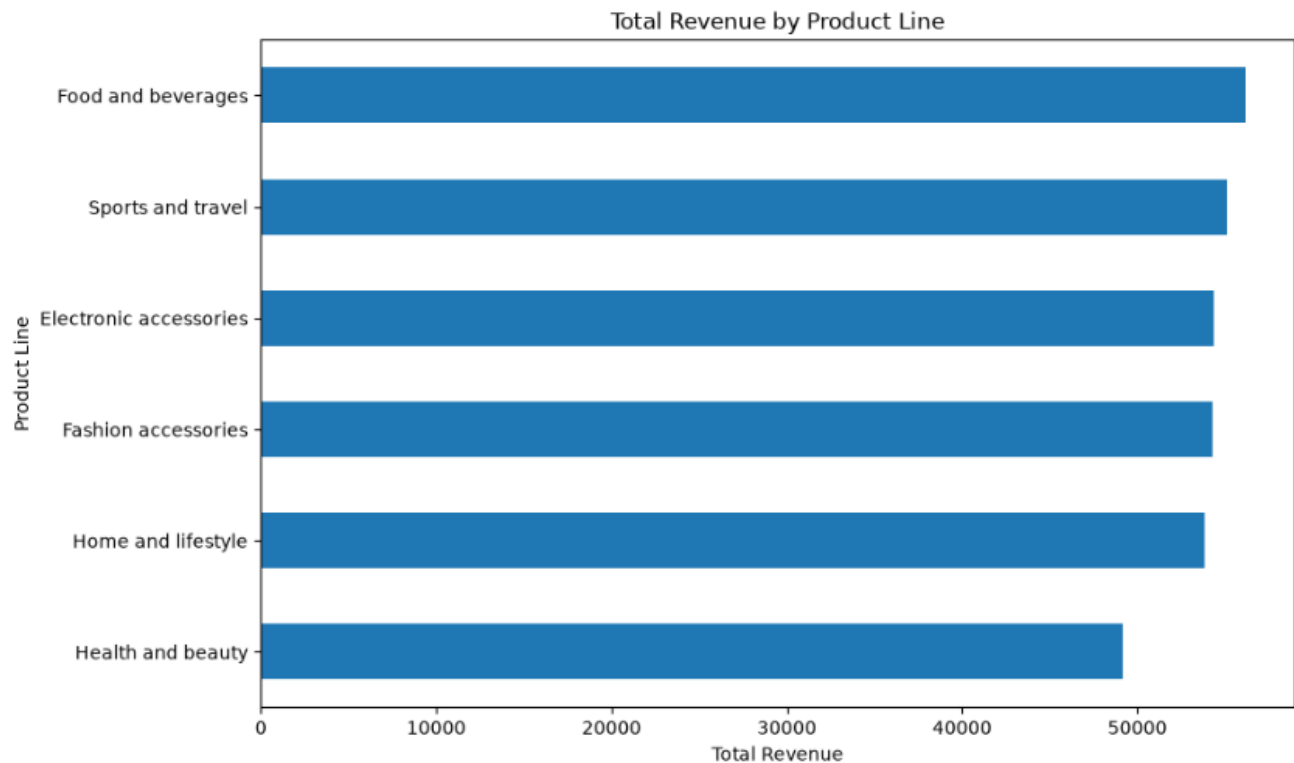
  product_line  unit_cost  quantity  5pct_markup  revenue \
0  Health and beauty    74.69         7    26.1415  548.9715
1  Electronic accessories  15.28         5     3.8200   80.2200
2  Home and lifestyle    46.33         7    16.2155  340.5255
3  Health and beauty    58.22         8    23.2880  489.0480
4  Sports and travel     86.31         7    30.2085  634.3785
5  Electronic accessories  85.39         7    29.8865  627.6165
6  Electronic accessories  68.84         6    20.6520  433.6920
7  Home and lifestyle    73.56        10    36.7800  772.3800
8  Health and beauty    36.26         2     3.6260   76.1460
9  Food and beverages    54.84         3     8.2260  172.7460
...
```

Fashion accessories

|

+-- High Revenue (Probability = 0.472, Payoff = 512.33)

+-- Low Revenue (Probability = 0.528, Payoff = 119.90)

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings...](#)

20. Highest Probability of High Revenue: Health and beauty

Probability: 0.5328947368421053

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Part B: Apply Expected Value**CODE:-****# 21. Explanation**

print(""" 21. Expected Value helps a business compare different product lines by considering both the probability of each possible revenue outcome and its payoff. The product line with the highest Expected Value is generally preferred. """)

22. Mathematical formula

```
print("\n22. Expected Value Formula:")
print("EV = Σ (Probability × Payoff)")
```

23. Calculate EV for one product line

```
example_product = decision_alternatives[0]
example_ev = (
    revenue_probability.loc[example_product, "High Revenue"]
    * revenue_payoff.loc[example_product, "High Revenue"]
    +
    revenue_probability.loc[example_product, "Low Revenue"]
    * revenue_payoff.loc[example_product, "Low Revenue"]
)
print("\n23. Expected Value of", example_product, "=", example_ev)
```

24. Calculate EV for every product line

```
expected_values = (
    revenue_probability * revenue_payoff
).sum(axis=1)
print("\n24. Expected Value of Every Product Line:")
print(expected_values)
```

25. Create final Expected Value DataFrame

```
ev_df = pd.DataFrame({
    "Product Line": decision_alternatives,
    "High Revenue Probability":
        revenue_probability["High Revenue"].values,
    "High Revenue Payoff":
        revenue_payoff["High Revenue"].values,
    "Low Revenue Probability":
        revenue_probability["Low Revenue"].values,
    "Low Revenue Payoff":
        revenue_payoff["Low Revenue"].values,
    "Expected Value":
        expected_values.values
})
print("\n25. Expected Value DataFrame:")
print(ev_df)
```

26. Compare Expected Values

```
print("\n26. Expected Value Comparison:")
print(
    ev_df[
```

```

["Product Line", "Expected Value"]
].sort_values(
    "Expected Value",
    ascending=False
)
)

```

27. Product line with highest Expected Value

```

highest_ev_product = expected_values.idxmax()
highest_ev = expected_values.max()
print("\n27. Highest Expected Value:")
print(highest_ev_product)
print("Expected Value:", highest_ev)

```

28. Product line with lowest Expected Value

```

lowest_ev_product = expected_values.idxmin()
lowest_ev = expected_values.min()
print("\n28. Lowest Expected Value:")
print(lowest_ev_product)
print("Expected Value:", lowest_ev)

```

29. Rank all product lines

```

ranking = expected_values.sort_values(
    ascending=False
)
print("\n29. Ranking from Highest to Lowest Expected Value:")
print(ranking)

```

30. Bar chart comparing Expected Values

```

plt.figure(figsize=(10, 6))
ranking.plot(kind="bar")
plt.title("Expected Value by Product Line")
plt.xlabel("Product Line")
plt.ylabel("Expected Value")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```

31. Compare highest High Revenue probability

```

# with highest Expected Value
print("\n31. Comparison:")
print(
    "Highest High Revenue Probability:",
    highest_high_probability_product
)
print(
    "Highest Expected Value:",
    highest_ev_product
)

```

32. Check whether they are the same

```

same_product = (
    highest_high_probability_product == highest_ev_product
)
print("\n32. Same Product Line?", same_product)

```

33. Change probability and recalculate EV

```

test_product = highest_ev_product
original_high_probability = revenue_probability.loc[
    test_product, "High Revenue"
]
original_low_probability = revenue_probability.loc[
    test_product, "Low Revenue"
]
high_payoff = revenue_payoff.loc[
    test_product, "High Revenue"
]
low_payoff = revenue_payoff.loc[
    test_product, "Low Revenue"
]

```

```

# Change High Revenue probability to 0.60

```

```

new_high_probability = 0.60
new_low_probability = 1 - new_high_probability
new_ev_probability = (
    new_high_probability * high_payoff
    +
    new_low_probability * low_payoff
)
print("\n33. Probability Change:")
print("Original High Revenue Probability:",
    original_high_probability)
print("New High Revenue Probability:",
    new_high_probability)
print("New Expected Value:",
    new_ev_probability)

```

34. Change payoff and recalculate EV

```

new_high_payoff = high_payoff + 50
new_ev_payoff = (
    original_high_probability * new_high_payoff
    +
    original_low_probability * low_payoff
)
print("\n34. Payoff Change:")
print("Original High Revenue Payoff:",
    high_payoff)
print("New High Revenue Payoff:",
    new_high_payoff)
print("New Expected Value:",

```

```
new_ev_payoff)
```

35. Automatic Expected Value calculation

```
def calculate_expected_values(data):
    counts = pd.crosstab(
        data["product_line"],
        data["revenue_level"]
    )
    probabilities = counts.div(
        counts.sum(axis=1),
        axis=0
    )
    payoffs = data.pivot_table(
        index="product_line",
        columns="revenue_level",
        values="revenue",
        aggfunc="mean"
    )
    ev = (
        probabilities["High Revenue"]
        * payoffs["High Revenue"]
        +
        probabilities["Low Revenue"]
        * payoffs["Low Revenue"]
    )
    best_product = ev.idxmax()
    return ev, best_product

automatic_ev, best_product = calculate_expected_values(df)
print("\n35. Automatic Expected Values:")
print(automatic_ev)
print("\nBest Product Line:")
print(best_product)

# 36. Display final decision tree with Expected Value
print("\n36. FINAL DECISION TREE")
print("=" * 70)
for product in decision_alternatives:
    print(f"\n[{product}]")
    print("  |")
    high_p = revenue_probability.loc[
        product, "High Revenue"
    ]
    high_pay = revenue_payoff.loc[
        product, "High Revenue"
    ]
    low_p = revenue_probability.loc[
        product, "Low Revenue"
    ]
    low_pay = revenue_payoff.loc[
        product, "Low Revenue"
    ]
    ]
```

```

product_ev = expected_values.loc[product]
print(
    f"    +--- High Revenue "
    f"(P={high_p:.3f}, Payoff={high_pay:.2f})"
)
print(
    f"    +--- Low Revenue "
    f"(P={low_p:.3f}, Payoff={low_pay:.2f})"
)
print(f"    Expected Value = {product_ev:.2f}")

```

37. Business Recommendation

```

print("\n37. BUSINESS RECOMMENDATION")
print(
    f"The business should select the '{highest_ev_product}' "
    f"product line because it has the highest Expected Value "
    f"of {highest_ev:.2f}. This indicates that, considering "
    f"both revenue probabilities and payoffs, it provides "
    f"the best expected financial outcome among the product lines."
)
print("KHUSHI T032")

```

OUTPUT:-

```

21. Expected Value helps a business compare different
product lines by considering both the probability of each
possible revenue outcome and its payoff.
The product line with the highest Expected Value is
generally preferred.

```

```

22. Expected Value Formula:
EV = Σ (Probability × Payoff)

```

```

23. Expected Value of Health and beauty = 323.6430197368422

```

```

24. Expected Value of Every Product Line:

```

```

product_line
Electronic accessories    319.632538
Fashion accessories       305.089298
Food and beverages        322.671517
Health and beauty         323.643020
Home and lifestyle         336.636956
Sports and travel          332.065220
dtype: float64

```

```

25. Expected Value DataFrame:

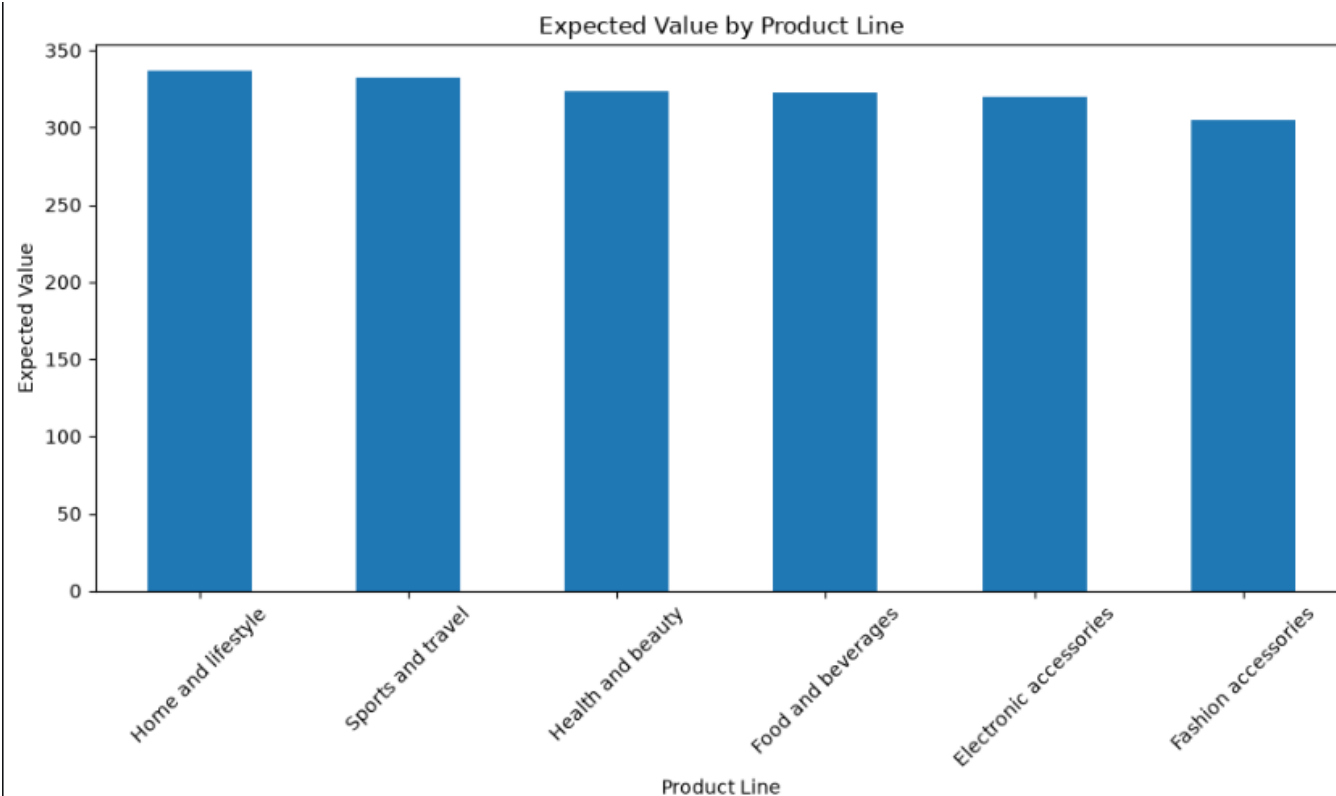
```

```

      Product Line  High Revenue Probability  High Revenue Payoff \
...
Food and beverages    322.671517
Electronic accessories  319.632538
Fashion accessories    305.089298
dtype: float64

```

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31. Comparison:

Highest High Revenue Probability: Health and beauty

Highest Expected Value: Home and lifestyle

32. Same Product Line? False

33. Probability Change:

Original High Revenue Probability: 0.5125

New High Revenue Probability: 0.6

New Expected Value: 371.2682503752345

34. Payoff Change:

Original High Revenue Payoff: 529.5827378048781

New High Revenue Payoff: 579.5827378048781

New Expected Value: 362.26195625

35. Automatic Expected Values:

product_line

Electronic accessories 319.632538

Fashion accessories 305.089298

Food and beverages 322.671517

Health and beauty 323.643020

Home and lifestyle 336.636956

Sports and travel 332.065220

...

37. BUSINESS RECOMMENDATION

The business should select the 'Home and lifestyle' product line because it has the highest Expected Value of 336.636956

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